SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

*I **KESHAVARAM L /192511253** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:
p = 17, q = 19

Compute n and ϕ(n)
Choose e = 5 and calculate the private key d
Encrypt the message M = 7
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

ANSWER:

RSA Key Generation and Security Analysis

Given:

- **p = 17**
- **q = 19**
- **e = 5**
- **Message M = 7**

Step 1: Compute n and ϕ(n)

$$\boldsymbol{n = p \times q = 17 \times 19 = 323}$$
$$\boldsymbol{\phi(n) = (17 - 1)(19 - 1) = 16 \times 18 = 288}$$

Step 2: Find Private Key (d)

Need

$$\boldsymbol{5d \equiv 1 \pmod{288}}$$

Using the Extended Euclidean Algorithm,

$$\boldsymbol{d = 173}$$

Public Key: (5,323)

Private Key: (173,323)

Step 3: Encrypt the Message

Formula:

$$\boldsymbol{C = M^e \bmod n}$$
$$\boldsymbol{C = 7^5 \bmod 323}$$
$$\boldsymbol{7^2 = 49}$$
$$\boldsymbol{7^4 = 49^2 = 2401 \equiv 140 \pmod{323}}$$
$$\boldsymbol{7^5 = 140 \times 7 = 980 \equiv 11 \pmod{323}}$$

Ciphertext = 11

Step 4: Decrypt

$$\boldsymbol{M = C^d \bmod n}$$
$$\boldsymbol{M = 11^{173} \bmod 323 = 7}$$

Original message is recovered.

Security Analysis

- Small prime numbers make factorization easy.
- An attacker can quickly compute p and q.
- Once p and q are known, the private key can be calculated.
- Therefore, RSA becomes insecure.
- Practical RSA uses 2048-bit or larger keys.

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2. ElGamal Encryption Analysis

Given:
p = 23, g = 5, private key x = 7, random k = 3, message M = 10

Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

ANSWER:

ElGamal Encryption Analysis

Given

- **p = 23**
- **g = 5**
- **x = 7**
- **k = 3**
- **M = 10**

Step 1: Public Key

$$
\begin{aligned}
&y = g^x \bmod p \\
&y = 5^7 \bmod 23 \\
&5^2 = 25 \equiv 2 \\
&5^4 = 2^2 = 4 \\
&5^7 = 5^4 \times 5^2 \times 5 \\
&= 4 \times 2 \times 5 = 40 \equiv 17
\end{aligned}
$$

Public Key:

$$(p, g, y) = (23, 5, 17)$$

Step 2: Encryption

$$
\begin{aligned}
&C_1 = g^k \bmod p \\
&C_1 = 5^3 = 125 \equiv 10
\end{aligned}
$$

Next,

$$
\begin{aligned}
&y^k = 17^3 \bmod 23 \\
&17^2 = 289 \equiv 13 \\
&17^3 = 13 \times 17 = 221 \equiv 14
\end{aligned}
$$

Now,

$$
\begin{aligned}
&C_2 = M \times y^k \bmod 23 \\
&10 \times 14 = 140 \\
&140 \equiv 2
\end{aligned}
$$

Ciphertext:

$$(C_1, C_2) = (10, 2)$$

Step 3: Decryption

Compute

$$\begin{aligned}
&C_1^x = 10^7 \mod 23 \\
&10^2 = 8 \\
&10^4 = 18 \\
&10^7 = 18 \times 8 \times 10 = 1440 \equiv 14
\end{aligned}$$

Inverse of 14 mod 23 = 5

$$\begin{aligned}
&M = C_2 \times 14^{-1} \mod 23 \\
&= 2 \times 5 = 10
\end{aligned}$$

Original message recovered.

Security Analysis

- Random value k changes every encryption.
- The same message generates different ciphertexts.
- Prevents pattern analysis.
- Improves confidentiality.

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3. RSA Digital Signature Verification

Given:
p = 11, q = 17, e = 7, message M = 8

Compute n and ϕ(n)
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

ANSWER:

RSA Digital Signature Verification

Given

- **p = 11**
- **q = 17**
- **e = 7**
- **M = 8**

Step 1: Compute n and ϕ(n)

$$\boldsymbol{n = 11 \times 17 = 187}$$
$$\boldsymbol{\phi(n) = 10 \times 16 = 160}$$

Step 2: Private Key

Need

$$\boldsymbol{7d \equiv 1 \pmod{160}}$$
$$\boldsymbol{d = 23}$$

Step 3: Generate Signature

$$\boldsymbol{S = M^d \bmod n}$$
$$\boldsymbol{S = 8^{23} \bmod 187}$$

Using modular exponentiation,

$$\boldsymbol{S = 151}$$

Digital Signature = 151

Step 4: Verification

$$\boldsymbol{M = S^e \bmod n}$$
$$\boldsymbol{151^7 \bmod 187 = 8}$$

The recovered message equals the original message.

Hence, the signature is valid.

Non-Repudiation Analysis

- Only the private key owner can generate the signature.
- Anyone can verify it using the public key.
- Prevents the sender from denying the message.
- Ensures authenticity and integrity.

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4. Diffie-Hellman with Attack Resistance Evaluation
Given:
p = 29, g = 2
User A private key = 5
User B private key = 12

Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

ANSWER:

Diffie–Hellman Key Exchange

Given

- **p = 29**
- **g = 2**
- **User A private key = 5**
- **User B private key = 12**

Step 1: Public Keys

For User A

$$\boldsymbol{A = 2^5 \ \mathrm{mod}\ 29}$$
$$\boldsymbol{A = 32 \equiv 3}$$

For User B

$$\boldsymbol{B = 2^{12} \ \mathrm{mod}\ 29}$$
$$\boldsymbol{2^{12} = 4096}$$
$$\boldsymbol{4096 \equiv 7}$$

Public Keys

A = 3

B = 7

Step 2: Shared Secret

User A computes

$$\mathbf{7^5 \bmod 29 = 16}$$

User B computes

$$\mathbf{3^{12} \bmod 29 = 16}$$

Shared Secret Key = 16

Attack Analysis

If an attacker replaces the exchanged public keys, each user unknowingly establishes a key with the attacker instead of each other (a Man-in-the-Middle attack). The attacker can decrypt, modify, and re-encrypt communications without either party noticing.

Solution

Use Digital Certificates, RSA Digital Signatures, or authenticated key exchange protocols (such as authenticated Diffie–Hellman) to verify public keys and ensure authenticity.

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5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500

Calculate total time taken for encryption
Compare with using only RSA for encrypting all data
Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

ANSWER:

Hybrid Encryption System Analysis

Given

- **RSA key exchange = 5 ms**
- **AES encryption = 0.5 ms/block**
- **Number of blocks = 500**

Step 1: Hybrid Encryption Time

AES time

$$\mathbf{500 \times 0.5 = 250\ ms}$$

Total Time

$$\mathbf{250 + 5 = 255\ ms}$$

Hybrid Encryption Time = 255 ms

Step 2: RSA Only

$$\mathbf{500 \times 5 = 2500\ ms}$$

RSA Only Time = 2500 ms

Step 3: Comparison

Time saved

$$\mathbf{2500 - 255 = 2245\ ms}$$

Hybrid encryption is approximately 10 times faster than using RSA alone for all data blocks.

Analysis

- **RSA securely exchanges the secret AES key.**
- **AES encrypts bulk data much faster than RSA.**
- **The combination provides both strong security and high performance.**
- **This approach is used in modern secure communication protocols such as HTTPS, TLS, VPNs, and secure messaging applications.**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

		engineering relevance			